

INTRODUCTION TO MACHINE LEARNING

CLUSTERING TECHNIQUES

WORK 3

December 8, 2024

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1 Introduction

This project evaluates the performance of various clustering techniques, including OPTICS (density-based), Spectral Clustering (graph-based), K-Means (partitioning), and Generalized Fuzzy C-Means, across three datasets from the UCI Machine Learning Repository: Hepatitis, MX, and Breast-w.

The workflow involves dataset selection, preprocessing to address missing values, and necessary transformations. Clustering algorithms will include sklearn-based implementations for OPTICS and Spectral Clustering, alongside custom implementations of K-Means, its improved versions (K-Means++ for better initialization and X-Means for automatic cluster detection), and an enhanced fuzzy clustering method.

Performance will be evaluated using internal metrics (cohesion and separation) and external metrics (alignment with true labels), focusing on the quality and relevance of the formed clusters.

2 Datasets

In this study, three datasets are analyzed without splitting into training and testing sets, allowing clustering algorithms to be applied directly for a comprehensive evaluation of their performance in grouping similar instances.

2.1 Hepatitis

The Hepatitis dataset contains 155 instances and 19 attributes, including six numerical ones: patient age, bilirubin levels, alkaline phosphatase, SGOT levels, prothrombin time, and albumin levels. The remaining 13 attributes are categorical with binary values yes or no, covering clinical symptoms and treatment indicators such as gender, steroid use, antiviral treatment, and symptoms like fatigue, anorexia, and enlarged liver. The target class has two outcomes: DIE and LIVE. The dataset shows an imbalanced class distribution with 79.35% in the LIVE class (Maj.Cla.) and 20.65% in the DIE class, resulting in a class distribution deviation of 29.35%. It has a 6% of missing values across attributes excluding age, sex, antivirals, and histology.

2.2 Mx

The Mx dataset comprises 2,048 instances, each described by 11 binary attributes labeled a00 to a10, with possible values of 0 or 1. The target variable, labeled "class," is also binary, taking values of 0 or 1. A key feature of this dataset is its perfectly balanced class distribution, with no deviation (Dev.Cla. = 0%). This means that 50% of the instances belong to one class, while the other 50% belong to the second class. Furthermore, the dataset is complete, containing no missing values.

2.3 Breast Cancer Wisconsin

The Breast Cancer Wisconsin dataset contains 699 instances, each characterized by 9 numeric attributes, with values ranging from 1 to 10. These attributes describe various cell characteristics, such as Clump Thickness, Cell Size Uniformity, and Mitoses. The target variable, *Class*, is categorical with two possible values: *benign* and *malignant*. The dataset's majority class accounts for 70.28% of instances, while the minority class represents 29.72%, resulting in a class deviation of 20.28%. Additionally, it contains 0.25% missing values across its attributes.

3 Data preprocessing

Data preprocessing cleans, normalizes, and transforms raw data to ensure compatibility with machine learning algorithms, enhancing accuracy, reliability, and insights.

For the **MX dataset**, minimal preprocessing was needed since all columns are binary and with no missing values. The primary step was decoding string values into numeric (0 and 1), ensuring compatibility with algorithms.

For the **Hepatitis dataset**, more extensive preprocessing was required due to mixed data types. Rows with over 30% missing values were removed, and numerical missing values were filled with the mean to maintain distribution. Categorical columns were decoded, mapped into binary form (e.g., “yes” to 1, “no” to 0), and remaining missing values were filled with the mode. Numerical columns were normalized using MinMaxScaler, scaling values between 0 and 1 for consistency and effective feature contribution.

For the **Breast Cancer Wisconsin dataset**, rows with over 30% missing values were removed, and remaining missing values in numeric columns were filled with the mean to preserve the overall distribution. All numerical attributes were normalized using MinMaxScaler, scaling their values between 0 and 1 to ensure consistency and balanced feature contribution. The target variable was decoded and mapped to binary values, with “benign” represented as 0 and “malignant” as 1.

Finally, the “class” column was dropped from all datasets to prepare for clustering, but it was saved separately for evaluating model performance later.

4 Analysis of clustering techniques

In the following sections, we explore several clustering methods: OPTICS, spectral clustering, k-means with two improved versions, and fuzzy clustering. Each technique is evaluated using both internal and external metrics to assess their performance.

We primarily used the **Silhouette Score** to evaluate the models, alongside other internal indices, depending on the clustering techniques. For external evaluation, we employed the **Purity Index**, which measures how well clusters contain only data points from a single class (ranging from 0 to 1, with higher values indicating better clustering). Additionally, we’ve used the **Jaccard Index**, which quantify the overlap between the clustering results and the true class assignments (ranging from 0 to 1, with values closer to 1 indicating better alignment), and also a pair counting metric as the **Rand Index** (from 0 to 1, better when close to 1) or the **Adjusted Rand Index** (from -1 to 1, better when close to 1).

5 Density-based: Optics Clustering

The Optics (Ordering Points to Identify the Clustering Structure) clustering algorithm is a density-based method designed to identify clusters of varying densities and discover structure in spatial datasets. It defines clusters as areas of high point density separated by areas of lower density. Points are classified into three categories: **core points**, which have the highest density; **border points**, which are less dense but near a core point; and **noise**, which consists of points with densities too low to belong to any cluster. Optics computes a “reachability distance” for each point, representing how closely it is connected to a core point.

In order to assess the quality of the clustering we’ve implemented a metric named **Connectivity**, which consists on computing a pairwise distance matrix and use it to identify the k-nearest neighbors for each point. Additionally, some parameters were set according to the datasets nature:

- **min_samples**: Minimum number of points needed to define a core point. Use smaller values (e.g., 5-10) for dense datasets and larger values for sparse ones.
- **cluster_method** and **xi**: The method xi handles varied density clusters. Smaller xi detects more clusters and fewer noise points, while larger xi merges nearby clusters.
- **min_clusters_size**: Clusters smaller than 2% of the dataset size are considered invalid.

Once having these parameters configured, we’ve experimented with several distance metrics [euclidean, cosine, l1] and algorithms [ball_tree, brute] to compute the nearest neighbors.

5.1 Hepatitis Dataset

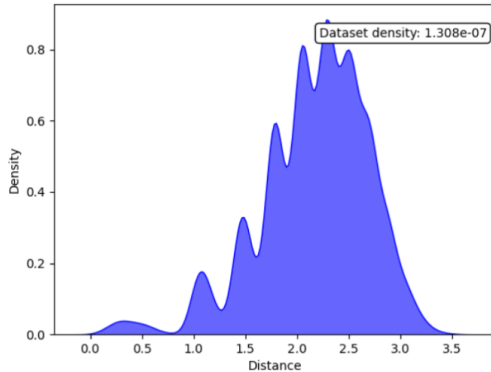


Figure 1: Hepatitis dataset density plot

For this dataset, the *min_samples* parameter was set to 15. Given that the dataset is mixed, containing both numerical and binary data, the distribution of pairwise distances reaches its peak around 2-2.5, which suggests a relatively high value and may indicate sparsity. The *xi* parameter was set to 0.001, as the algorithm had difficulty identifying clusters and tended to classify most points as noise.

With this configuration, OPTICS successfully identified **two clusters** using the **cosine** distance metric and the **brute force** algorithm, with densities of 2.433×10^{-7} and 5.332×10^{-8} , respectively. Out of 155 total points, 49 were classified as noise. The clustering achieved a moderate **Silhouette Score** of 0.414 and a **Connectivity** value (better when close to 0, measures how well points within the same cluster are connected to each other based on the overall structure of the data) of 0.4966. In terms of external metrics, it attained a **Purity** of 0.764, and a **Jaccard Index** of 0.513. Additionally, the reachability plot can be visualized for further insights into the clustering process performed by the algorithm.

The valleys in Figure 2 correspond to the clusters, with deeper valleys indicating higher density. The clusters are represented by relatively flat regions with low reachability distance, which signifies dense areas. Black points between the two clusters represent noise points, indicating regions with insufficient density for cluster assignment. Cluster 1 (brown) has a higher reachability distance compared to Cluster 2 (blue), suggesting that it is less dense overall.

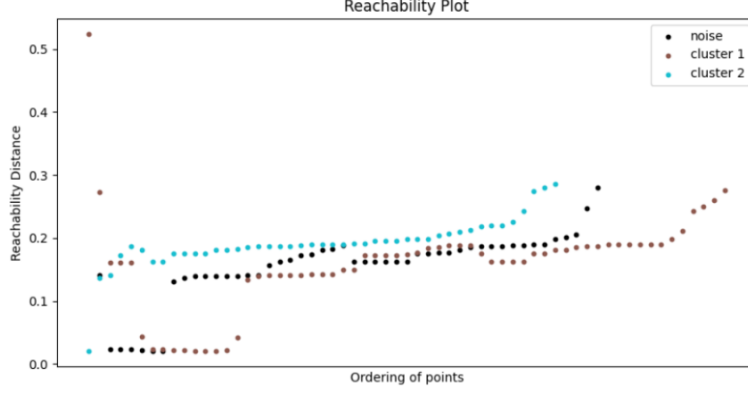


Figure 2: Hepatitis dataset reachability plot

5.2 Mx Dataset

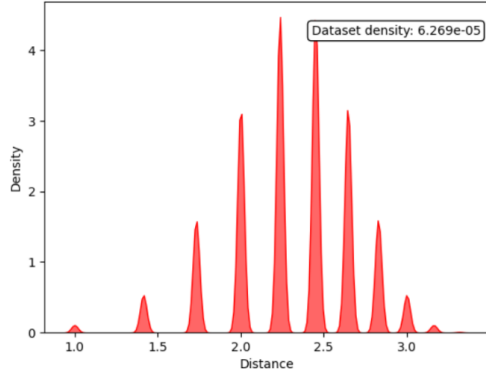


Figure 3: Mx dataset density plot

This dataset appears to be sparse but more uniformly distributed. For this reason, *min_samples* was set to 5, and *xi* was increased to 0.05. With a dataset size of 2048, the algorithm struggled with the clustering task. For both Euclidean and L1 distance metrics, it neither created clusters nor classified noise points. With the cosine distance metric, the algorithm only managed to classify 1981 points as noise. Same results were achieved even with varying parameters (to be a bit more flexible).

5.3 Breast Cancer Dataset

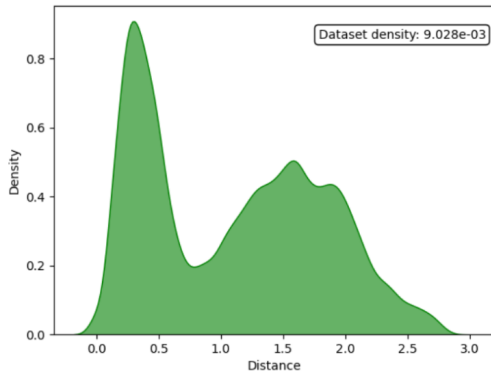


Figure 4: Breast dataset density plot

This final dataset is clearly denser than the previous ones, which is why *min_samples* was set to 3 and *xi* to 0.001. In this case, the algorithm successfully formed clusters for all combinations of parameters. For both Euclidean and L1 distances, it formed 5 clusters, classifying 576 and 584 noise points, respectively, out of a total of 699. While the **Silhouette Score** yielded very good results, **Connectivity** achieved disastrous ones, suggesting that the clustering method may prioritize separation and compactness over neighborhood relationships, failing to maintain spatial or logical continuity. On the other hand, for cosine distance, the algorithm created 8 clusters and classified 496 noise points. In this case, **Silhouette Scores** are moderate, while **Connectivity** remains poor.

Distance Metric	Algorithm	Silhouette Score	Connectivity	# Clusters	# Noise Points
euclidean	ball_tree	0.922	0.052	5	576
euclidean	brute	0.922	0.052	5	576
cosine	brute	0.566	0.063	8	496
l1	ball_tree	0.94	0.023	5	584
l1	brute	0.94	0.023	5	584

Table 1: Performance metrics for different distance metrics and algorithms.

Table 1 summarizes these results mentioned above. It is important to notice that column values do not vary between the same distance metric. To investigate this further, we computed the Hamming Distance between the labels produced by each of these combinations.

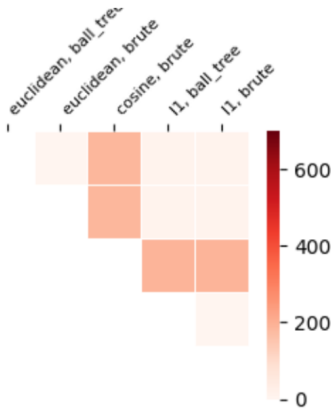


Figure 5: Hamming matrix plot

Note that notable changes in label assignment occur when changing the distance metric from Euclidean or L1 to Cosine. Moreover, within the same distance metric, changing the algorithm to compute nearest neighbors does not have a significant impact. In general, and also considering the previous datasets, we can conclude that while Euclidean and L1 distances perform similarly, Cosine acts more independently, producing very different results. This is logical, as the first two focus more on the magnitudes of differences between data points, while the latter focuses on angular relationships. Consequently, if the dataset contains features where magnitudes vary greatly, Cosine distance will prioritize orientation, causing its clustering results to diverge from those based on spatial measures.

6 Graph-based: Spectral Clustering

Spectral clustering is a powerful technique for identifying clusters with non-convex shapes. It relies on the eigenstructure of a similarity matrix, which encodes pairwise similarities between data points. In our analysis, we explored two methods to compute this matrix: the Gaussian (RBF) kernel and the k-nearest neighbors method. The similarity matrix is transformed into a graph Laplacian, which captures the structure of the underlying graph. Eigenvectors corresponding to the smallest eigenvalues of the Laplacian are then computed using either the ARPACK or LOBPCG method, both optimized for large, sparse, and symmetric matrices. Finally, a clustering algorithm like k-means or cluster-qr is applied to assign labels.

The similarity matrix can also be viewed as a weighted graph, where nodes represent data points and edge weights represent similarities. To evaluate clustering quality, we used the **Modularity Metric**, which measures the density of edges within clusters versus between clusters, and the **Dunn Index**, which compares inter-cluster separation to intra-cluster compactness. For k-nearest neighbors, the adjacency matrix corresponds directly to the affinity matrix, while for the RBF kernel, it is thresholded. High **Modularity** and **Dunn Index** values indicate well-defined clusters.

Given the following set of parameters:

affinities: $\{nearest_neighbors, rbf\}$
 n_neighbors: $\{3, 7, 11\}$
 eigen_solvers: $\{arpack, lobpcg\}$
 assign_labels: $\{kmeans, cluster_qr\}$

we performed the analysis as follows:

1. For each affinity calculation technique and for different numbers of clusters (k), ranging from 2 to 6, we averaged the **Silhouette Score**, **Modularity Metric**, and **Dunn Index** across all combinations of *eigen_solvers* and *assign_labels*. This allowed us to analyze the impact of different affinity calculations on clustering performance.
2. Specifically, for the RBF affinity calculation, we examined whether varying the eigen solvers and label assignment strategies influenced the clustering results.

6.1 Hepatitis dataset

As shown in Figure 6, the results for the first dataset do not indicate a clear k value that consistently delivers the best performance. The **Silhouette Scores** generally peak at $k = 2$, while the **Dunn Index** tends to reach its highest values at $k = 3$ and rises again at $k = 6$. Regarding **Modularity**, it increases proportionally with k , reflecting stronger cluster structures as more clusters are formed. This behavior is expected, as smaller and more localized clusters tend to align better with regions of high edge density in the graph, thereby increasing **Modularity**.

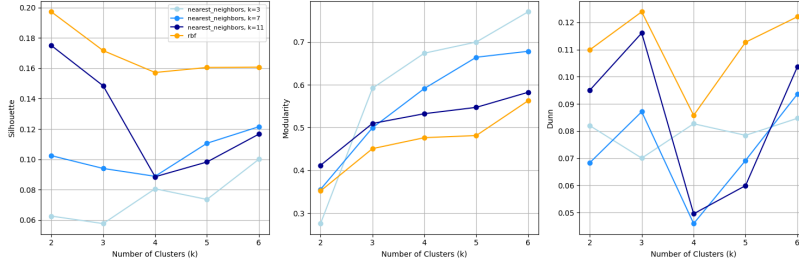


Figure 6: Spectral clustering internal indices varying k values for the Hepatitis dataset for different affinities

Setting RBF kernel for affinity calculation, Figure 7 shows only two line plots. This occurs because performance remains unchanged regardless of the eigen solver, as lines overlap completely. Regarding label assignment, *k-means* clearly outperforms the *cluster-qr* method.

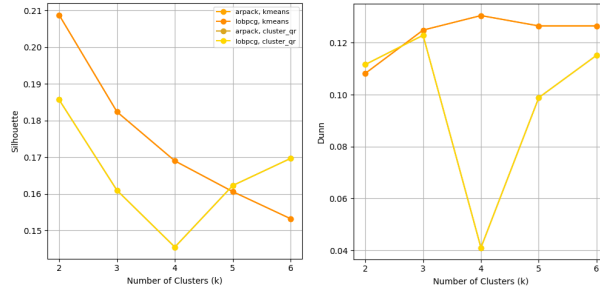


Figure 7: Spectral clustering internal indices varying k values for the Hepatitis dataset for different eigen solvers and label assignment methods

For the external evaluation, we assessed three models across the above specified number of clusters. The first two models used the Gaussian (RBF) kernel for affinity computation, while the third model relied on nearest neighbors with $k = 6$.

k	Affinity	N neighbors	Purity	Adjusted Rand Index	Jaccard Index
2	rbf	-	0.793	0.303	0.569
3	rbf	-	0.825	0.213	0.448
6	nearest neighbors	11	0.871	0.072	0.204

Table 2: External indices for the Hepatitis dataset

While the Purity Scores are relatively high across all rows, indicating that the clusters contain a large proportion of points from single true classes, the low **ARI** and **Jaccard Index** reveal poor overall alignment with the true class structure. Especially in the last model, the significant gap between **Purity** and **ARI** highlights a tendency for the model to over-cluster, splitting true classes into multiple smaller clusters. Among the models, the first one achieved the best clustering performance by striking a better balance across all scores.

6.2 Mx dataset

For the Mx dataset, it is evident that two clusters consistently achieve the highest values for both the **Silhouette Score** and the **Dunn Index** across all parameter combinations. However, several important observations can be drawn from the analysis presented in Figure 8. Notably, the **Modularity** values are significantly higher, indicating that the graph structure derived from the affinity matrix (computed using the nearest neighbors method) effectively captures the cluster relationships. On the other hand, it is important to highlight that the **Silhouette Score** drops to very low values. This suggests that the clusters are most likely non-convex in shape, as their structure does not align well with convexity assumptions inherent in the **Silhouette** metric.

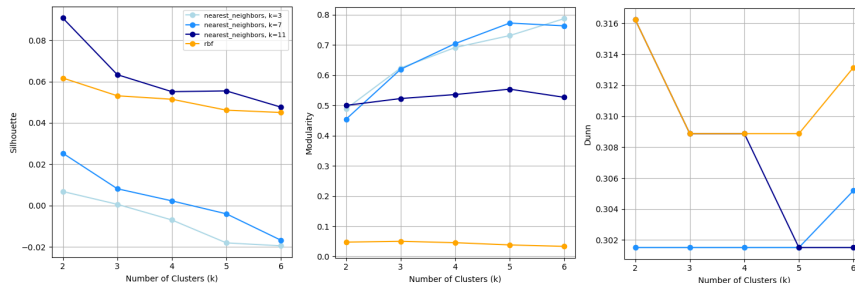


Figure 8: Spectral clustering internal indices varying k values for the Mx dataset for different affinities

From Figure 9, we can infer that the choice of eigen solver and label assignment method depends on the number of clusters. While using a smaller number of clusters ($k=2$), the cluster-qr method performs better, conversely, for larger values of k , the k-means approach is more suitable. Once again, it is evident that the choice of eigen solver has minimal impact on the results.

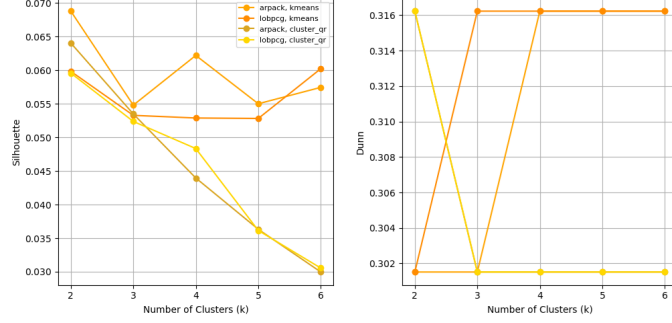


Figure 9: Spectral clustering internal indices varying k values for the Mx dataset for different eigen solvers and label assignment methods

For the evaluation using external indices, we tested two different affinity computation methods with $k=2$, as shown in Table 3. Although these configurations yield higher **Modularity** scores, the resulting clusters are poorly defined and show weak alignment with the ground truth labels.

k	Affinity	N neighbors	Purity	Adjusted Rand Index	Jaccard Index
2	rbf	-	0.557	0.012	0.339
3	rbf	-	0.526	0.002	0.251
6	nearest neighbors	11	0.571	0.008	0.162

Table 3: External indices for the Mx dataset

6.3 Breast dataset

Figure 10 shows that the Breast dataset exhibits a strong clustering structure for $k = 2$, as evidenced by the highest **Silhouette Score**, except when using the nearest neighbors approach with 3 neighbors. This suggests that exploring higher values for the nearest neighbors parameter could be beneficial. Additionally, the RBF kernel model reaches its peak **Dunn Index** at $k = 3$, while also maintaining a high **Silhouette Score**.

Interestingly, the nearest neighbors approach with 11 neighbors yields high **Modularity Metric** values but relatively low **Dunn Indices**. This discrepancy may arise from clusters being elongated or irregularly shaped, which increases intra-cluster distances and results in a lower **Dunn Index**.

As with the other datasets, the **Modularity Metric** remains exceptionally low for the RBF kernel model. This likely indicates that the adjacency or similarity matrix used to construct the graph is poorly defined, leading to low scores in this metric. This matrix is generated by applying the RBF kernel to the pairwise Euclidean distances between points and then thresholding the resulting values. Experimenting with alternative graph construction methods could potentially improve the **Modularity Metric** and enhance its interpretability.

Figure 11 shows that varying the eigen solvers and label assignment methods with the RBF kernel model has no significant impact on clustering quality. The internal evaluations remain very similar, with peaks observed at $k = 2$ and $k = 3$.

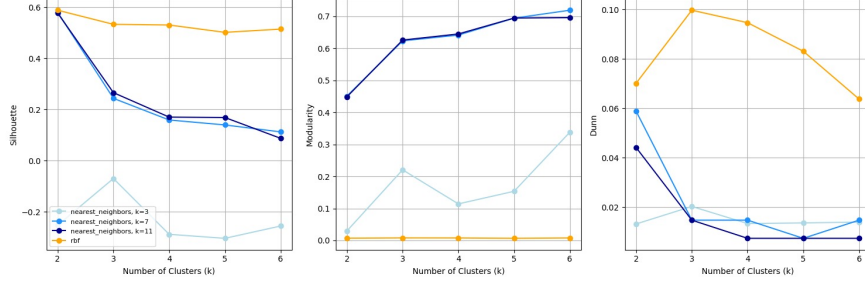


Figure 10: Spectral clustering internal indices varying k values for the Breast dataset for different affinities

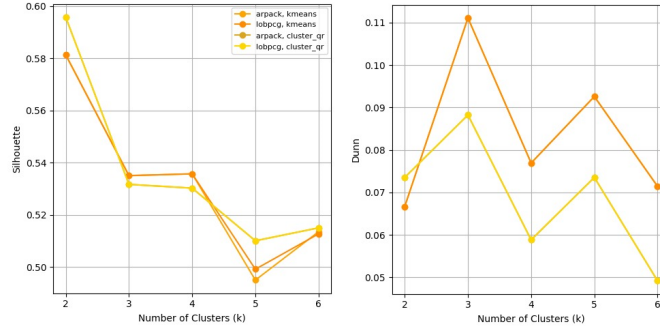


Figure 11: Spectral clustering internal indices varying k values for the Breast dataset for different eigen solvers and label assignment methods

Another indication of the elongated shape of the clusters in this dataset is the superior performance of the nearest neighbors model compared to the RBF kernel one, achieving an impressive score exceeding 0.9 in both **Purity** and **Jaccard Index** (see Table 4). This suggests that the clusters are not compact or well-separated by global distance metrics but are instead better defined by local proximity and connections within neighborhoods.

k	Affinity	N neighbors	Purity	Adjusted Rand Index	Jaccard Index
2	rbf	-	0.921	0.704	0.775
2	nearest neighbors	11	0.971	0.888	0.902
3	rbf	-	0.951	0.747	0.791

Table 4: External indices for the Breast dataset

7 K-Means

K-Means is a clustering algorithm that divides a dataset into k clusters by minimizing variance within each cluster. It starts with randomly initialized centroids, assigns data points to the nearest centroid, and recalculates centroids as the mean of their cluster points. This is repeated until centroids stabilize or a maximum number of iterations is reached. While efficient and simple, K-Means requires specifying k in advance and is sensitive to initial centroid placement, which can affect results.

7.1 Hepatitis Dataset

For the Hepatitis dataset, we evaluated the clustering performance of K-Means using three metrics: the **Silhouette score** and **Calinski-Harabasz Index** as internal indices, and the **Rand Index** as external. The **Calinski-Harabasz** measures the ratio of the sum of between-cluster dispersion to within-cluster dispersion. It has no upper bound, and higher values signify more distinct clusters. These metrics were calculated using Euclidean, Manhattan, and Cosine distance metrics, averaged over several runs.

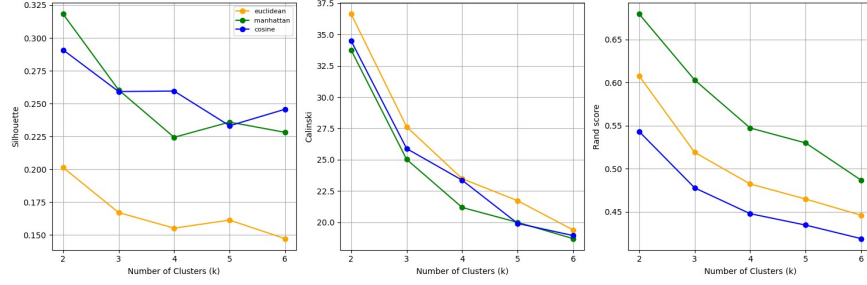


Figure 12: K-Means evaluation metrics varying k values for the Hepatitis dataset

From Figure 12 we can see that across all metrics and distance measures, the highest values consistently occur at $k=2$, aligning with the number of distinct classes in the dataset. Additionally, we can observe that as k increases, all metrics show a steady decline without a clear elbow point, indicating diminishing cluster quality for highest k values. Therefore, based on these observations, we conclude that the optimal number of clusters for this dataset is $k=2$, with the Manhattan distance generally performing the best.

7.2 Mx Dataset

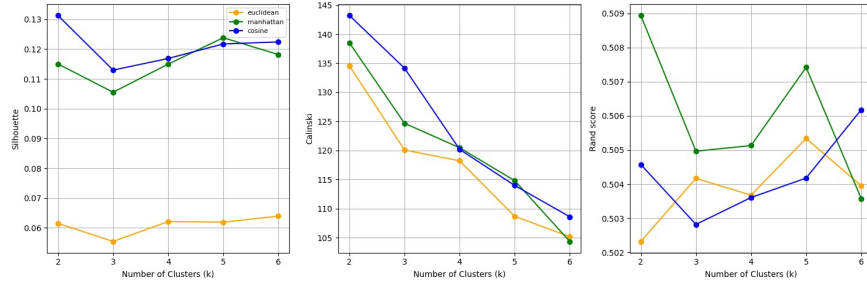


Figure 13: K-Means evaluation metrics varying k values for the Mx dataset

Figure 13 illustrates the clustering evaluation for the Mx dataset using the three aforementioned metrics. For the **Silhouette Score**, all distance metrics show consistently low values across all k -values, ranging from 0.06 to 0.14, with minimal variation (0.02) within each distance metric. While an elbow can be observed at $k=3$, where values slightly increase, the change is not significant. In contrast, the **Calinski-Harabasz Index** exhibits a clear downward trend, with lower values as k increases, suggesting that clustering performance deteriorates for higher k -values. Regarding the **Rand Index**, values show little variation between k -values, ranging from 0.502 to 0.509. Despite the lack of strong patterns in the **Silhouette Score** and **Rand Index**, the **Calinski-Harabasz**

Index clearly indicates that increasing k beyond 2 results in poorer clustering quality. Therefore, we select $k = 2$ as the optimal number of clusters for this dataset.

7.3 Breast Cancer Dataset

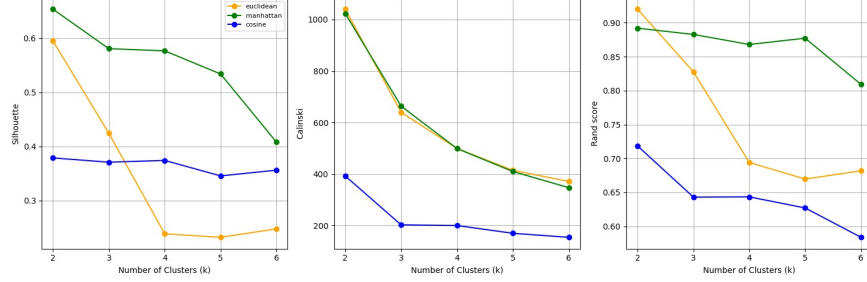


Figure 14: K-Means evaluation metrics varying k values for the Breast Cancer dataset

As with the previous two datasets, the Breast Cancer dataset was evaluated using the same metrics and distance measures. Figure 14 reveals a clear trend where clustering quality deteriorates as k increases, as indicated by the decline in all three metrics. Notably, $k = 3$ appears to be an elbow, with a more significant jump observed between $k = 2$ and $k = 3$. Beyond $k = 3$, the values change only slightly, with $k = 3$ showing considerably lower values compared to $k = 2$, and the metrics gradually diminishing for higher values of k . Given that, the results for $k = 2$ remain significantly higher across all metrics, we conclude this k as the optimal number of clusters, with Manhattan distance generally performing the best.

Finally, to better understand the clustering structure, we applied PCA to reduce the dimensionality of the data and visualized the clustering results using the Manhattan distance.

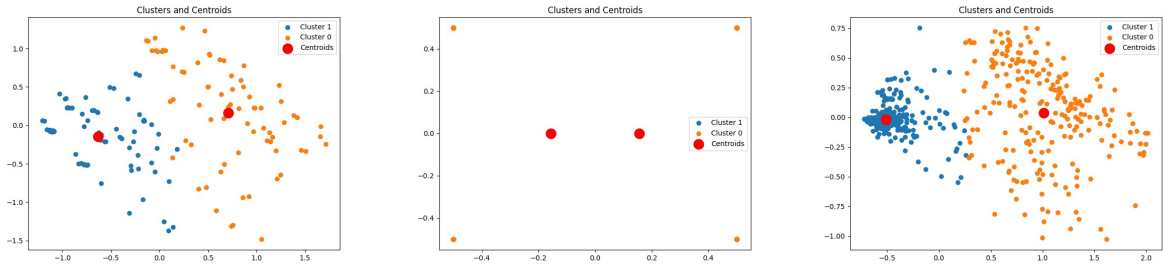


Figure 15: PCA visualization of clustering for $k = 2$ on the Hepatitis, Mx, and Breast Cancer datasets, respectively.

The left and right plots display well-separated clusters for the Hepatitis and Breast Cancer datasets, respectively, while the middle plot for the Mx dataset shows poor clustering results, primarily due to the all-categorical nature of the data, as previously highlighted by the Silhouette Score.

8 Improved K-Means

8.1 K-Means++

K-Means++ is an improved version of the standard K-Means algorithm. It addresses its sensitivity to initial centroid placement. Instead of random initialization, K-Means++ selects the first centroid randomly and subsequent centroids based on their distance from already chosen ones, with a probability proportional to the squared distance. This ensures evenly spread initial centroids, improving convergence and consistency.

8.2 X-Means

X-Means is an extension of the K-Means algorithm designed to address one of its primary limitations: the need to predefine the number of clusters, k . Instead, X-Means dynamically determines the optimal number of clusters during the clustering process. The algorithm begins with a small initial number of clusters, k_{min} , and iteratively evaluates whether splitting existing clusters improves the overall model. To make this decision, X-Means employs the **Bayesian Information Criterion (BIC)**, which balances model fit (how well the clusters represent the data) with model complexity (the number of clusters). A cluster is split only if the resulting BIC score improves, ensuring that new clusters are added only when they significantly enhance the model's quality. This dynamic approach makes X-Means more adaptive to the underlying structure of the data.

8.3 Analysis and Comparison

This section evaluates and compares the performance of the three clustering methods—KMeans, KMeans++, and X-Means—using both internal and external metrics. For consistency, the optimal number of clusters (k) and the distance metric, both determined in the previous K-Means section, were applied to all methods, as the enhanced variants (K-Means++ and X-Means) are based on the standard K-Means algorithm. Across all three datasets, the optimal k was 2, and the best-performing distance metric, *Manhattan*, was used.

Given the inherent randomness in these algorithms, each was executed 15 times per dataset, and the results presented in the plots reflect the average performance across these runs.

For this comparison, the **Silhouette Score** was used as an internal index, while the **Jaccard Index**, **Purity**, and **F1 Score** were used as external indices. As observed in the graphs in Figure 16, it cannot be affirmed that the improved K-Means algorithms, KMeans++ and X-Means, enhance the performance of standard K-Means, as in almost every graph and dataset, they achieve similar performance. The only noticeable variation is in the F1 Score, and not always with better results.

Furthermore, these graphs also illustrate the clustering performance depending on the dataset. In this case, it can be observed that the **Mx dataset** yields the worst results. This may be due to the nature of the dataset, which consists entirely of categorical values (0 and 1). Next, the **Hepatitis dataset** shows relatively poor clustering performance according to the internal Silhouette Metric. However, when considering the other three external metrics, which compare the labels assigned by the algorithms with the true labels, the performance is not as poor. This performance could also be attributed to the nature of the dataset, which is mixed, containing both numerical and categorical attributes. Lastly, the dataset with the best clustering performance is the **Breast Cancer dataset**, which achieves very high values in both the Jaccard Index and Purity. This strong clustering performance can likely be attributed to the fact that the dataset consists entirely of numerical attributes.

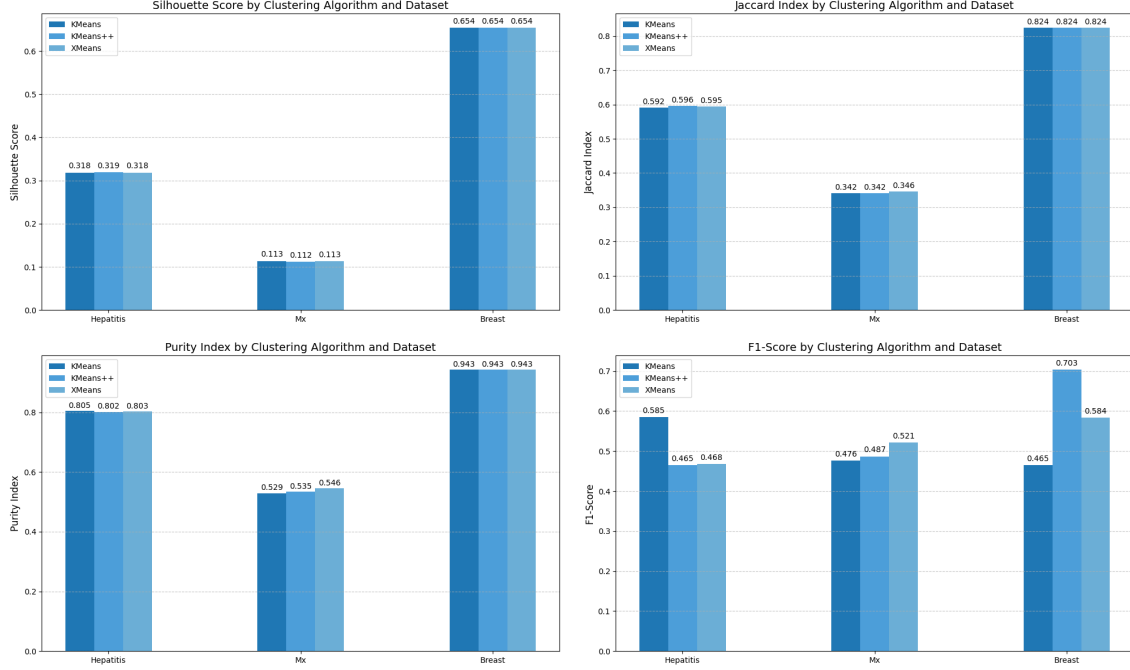


Figure 16: Comparison of clustering performance of K-Means, K-Means++.

In conclusion, it can be stated that there is no clear improvement in the enhanced K-Means algorithms (KMeans++ and X-Means) compared to standard K-Means. Additionally, the performance of these clustering algorithms varies depending on the nature of the dataset, with categorical-only datasets generally performing poorly and numerical-only datasets performing better.

9 Fuzzy Clustering

For the fuzzy clustering, we have chosen to implement the Generalized fuzzy C-means clustering algorithm with improved fuzzy partition, which is an enhanced version of the traditional Fuzzy C-Means (FCM) algorithm. This method addresses some of the limitations of the original FCM, particularly in handling noisy data and ensuring more accurate clustering in cases where data points may be poorly separated or contain outliers.

Similar to FCM, GIFP-FCM starts by specifying the number of clusters c and randomly initializing the membership matrix U , where each element indicates a data point's degree of belonging to a cluster. The fuzziness parameter m is also set to control membership fuzziness. The main difference in GIFP-FCM is the incorporation of an improved partition strategy that aims to refine the membership matrix and cluster centers. During each iteration, GIFP-FCM alternates between:

1. Updating the cluster centers, V , using a weighted average of the data points, similar to FCM, but with improvements that better reflect the true structure of the data (see formula 7 from [5]).
2. Refining the membership values, U , based on the updated cluster centers and adjusting the memberships to ensure a more precise partition of the data (see formula 4 from [5]).

The process continues iteratively until convergence, defined as minimal changes in cluster centers,

or when a maximum number of iterations is reached.

As can be seen in the formulas, this algorithm uses 3 parameters which need to be set: the number of clusters, k , the fuzzifier parameter m , and the suppression parameter, w . To simplify the analysis, we first fixed a value for the suppression parameter. To achieve this, we plotted all possible parameter combinations using the **Partition Entropy**, the metric that provided the most conclusive results. As shown in Figure 17, the **optimal suppression value** is **$w=0.99$** for all datasets. Smaller values tend to improve cluster differentiation by reducing the influence of distant data points.

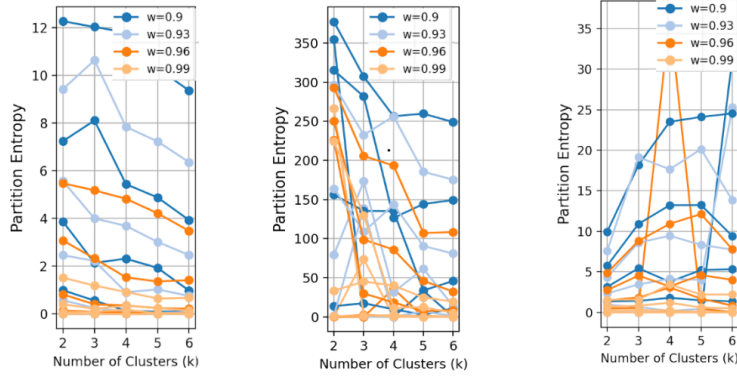


Figure 17: Partition Entropy for different suppression values across the Hepatitis, Mx, and Breast Cancer datasets, respectively.

9.1 Hepatitis Dataset

For each dataset, we created several plots that helped us to discuss which was the better value to use for each of them. In particular, we used the **Silhouette score**, the **Xie-Beni Index** (smaller values indicate better-defined clusters) and the **Partition Entropy** (lower values indicate more distinct clusters).

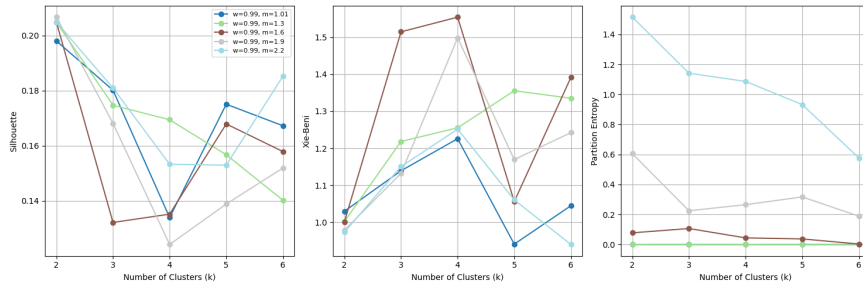


Figure 18: Fuzzy clustering internal indices varying k values for the Hepatitis dataset for different parameter combinations

At first glance, the **Silhouette plot** in Figure 18 does not conclusively indicate the optimal parameter m . However, it highlights $k=2, 5, 6$ as promising cluster numbers. This aligns with the **Xie-Beni plot**, where these k -values exhibit lower index values. The partition entropy plot reveals notable differences between m -values, showing that lower m -values result in lower entropy due to more distinct memberships and reduced uncertainty. Thus, the optimal model likely uses $k=2$ with a low m value such as $m=1.01, 1.3$. To finalize the choice, we will further examine these parameter

combinations and include $k=6$, $m=1.01$ for comparison, given its promising results aside from the partition coefficient.

k	m	Purity	Rand Index	Jaccard Index
2	1.01	0.793548	0.634185	0.535309
2	1.3	0.793548	0.620947	0.520737
6	1.01	0.819355	0.520737	0.204236

Table 5: External indices for the Hepatitis dataset

From table 5 we see that the model with $k=2$, $m=1.01$ stands out as the best, achieving the highest **Rand Index** and **Jaccard Index** with strong **Purity** (0.7935), indicating excellent clustering quality. While $k=2$, $m=1.3$ is a solid alternative, external metrics show lower values which indicates it might be less optimal. The model with $k=6$, $m=1.01$, despite having the highest **Purity** (0.819), shows poor alignment with the ground truth, with a much lower **RI** and **Jaccard Index** (0.20). Thus, $k=2$, $m=1.01$ is the most reliable choice.

9.2 Mx Dataset

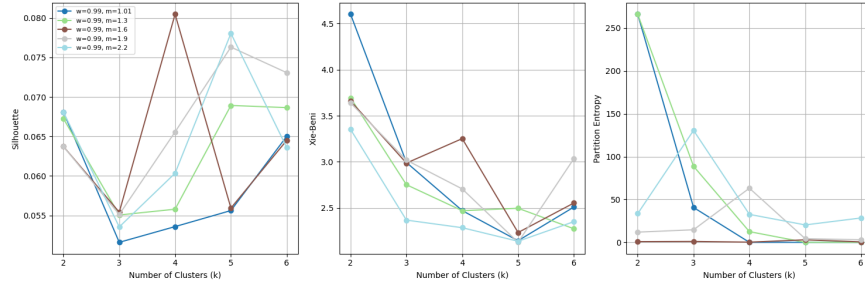


Figure 19: Fuzzy clustering internal indices varying k values for the Mx dataset for different parameter combinations

The **Silhouette** score for the Mx dataset (Figure 19) is very low, peaking at just 0.08, indicating poorly separated clusters. The **Xie-Beni Index** confirms this with high values reflecting weak cohesion and separation. Finally, the Partition Entropy plot shows an inverse relation between k and m values by giving better results for large k and small m or small k and bigger m values.

In summary, fuzzy clustering appears unsuitable for this dataset. However, based on external metrics, the best models for further analysis are $k=2, 4, 5$ and $m=1.01, 1.6, 2.2$.

k	m	Purity	Rand Index	Jaccard Index
2	1.01	0.504883	0.499803	0.333824
4	1.6	0.582031	0.508335	0.208289
5	2.2	0.546875	0.504152	0.183756

Table 6: External indices for the Mx dataset

The results in Table 6 reveal trade-offs between the evaluated metrics. The parameter combination $k=4$, $m=2.2$ achieves the highest **Purity** (0.5820) and **RI** (0.5083), indicating well-separated clusters, but sacrifices overlap coverage, as reflected in the lower **Jaccard Index**. Conversely, $k=2$, $m=1.01$ achieves the highest **Jaccard Index**, suggesting better coverage of true clusters, though

its **Purity** and **Rand Index** are slightly lower, implying less precise assignments. Finally, $k=6$, $k=1.01$ provides moderate performance in most of the metrics but has the lowest **Jaccard Index**, suggesting it over-partitions the data into smaller, less cohesive clusters. Overall, $k=4$, $m=2.2$ is ideal for precision and cluster cohesion, while $k=2$, $m=1.01$ is preferable for capturing broader overlaps, depending on the task’s priorities.

9.3 Breast Cancer Dataset

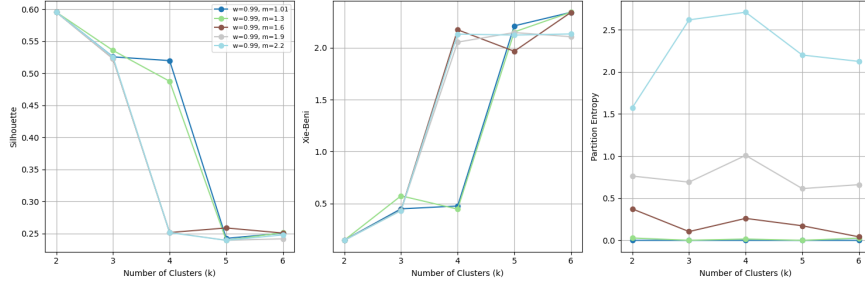


Figure 20: Fuzzy clustering internal indices varying k values for the Breast dataset for different parameter combinations

For the Breast dataset, the **Silhouette** plot in Figure 20 shows the best performance among the three datasets, with values reaching 0.6 when $k=2$ across all m values. This suggests that, while not perfectly separated, the clusters are better than random, effectively grouping similar points with reasonable separation. In contrast, the **Xie-Beni Index** for $k=6$ is notably high, indicating excessive clusters reduce compactness and coherence.

In conclusion, unlike the Mx dataset, fuzzy clustering works well on the Breast dataset, with the best results observed for $k=2,3,4$, and low m values like $m=1.01$.

k	m	Purity	Rand Index	Jaccard Index
2	1.01	0.959943	0.922985	0.869515
3	1.01	0.965665	0.893823	0.815224
4	1.01	0.962804	0.865629	0.767073

Table 7: External indices for the Breast dataset

From Table 7, we observe that all models perform really well, achieving high **Purity** (0.96), indicating accurate and well-aligned clustering with the true labels. However, we notice that as k increases, the overall behaviour gets worse. For instance, the model with $k=4$, $m=1.01$ gets almost 10% poorer values both for **RI** and **Jaccard Index** metrics, reflecting less accurate clustering.

10 Conclusions

The analysis of unsupervised clustering algorithms reveals that the ability to identify meaningful clusters without access to ground truth is inherently challenging. The performance of the clustering algorithms was heavily influenced by the internal characteristics of each dataset, with no single algorithm consistently outperforming the others. This highlights the complexity of clustering tasks and the need for careful consideration of data properties. As shown in Figure 21, all algorithms demonstrated comparable performance across different metrics.

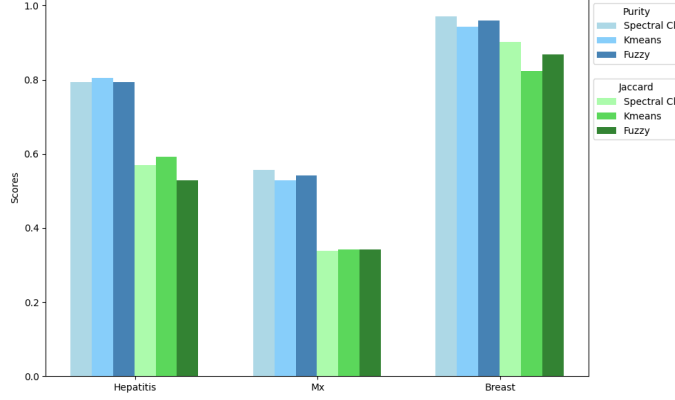


Figure 21: Final external evaluation of the best-performing models

- Numerical Datasets:** The algorithms performed best on numerical datasets, as demonstrated by the Breast dataset. In this case, the clusters were well-defined and effectively captured, with consistent results across all algorithms. The optimal number of clusters (k) identified for the Breast dataset was 2, a value agreed upon by most algorithms. As shown in Figure 21, the best-performing algorithm for this dataset was Spectral Clustering. Nevertheless, the spectral clustering results suggest that the clusters are likely non-convex, as indicated by discrepancies between the Modularity and Dunn Index, as well as the superior performance of the nearest neighbors model.
- Mixed Datasets:** For mixed datasets like the Hepatitis dataset, which combines categorical and numerical attributes, the algorithms faced challenges due to the dataset’s imbalance, small size (155 rows), and the disproportionate number of features (20). The best k for Hepatitis was consistently 2 across all algorithms, suggesting that a binary clustering structure is most suitable for this dataset. As seen in Figure 21, the algorithm which worked best with this dataset was K-means.
- Categorical Datasets:** The Mx dataset, which is entirely categorical, proved to be the most difficult for clustering algorithms. Although almost all algorithms agreed that $k = 2$ was the optimal number of clusters, the scores on the evaluation metrics were very low, indicating that the clusters formed were not representative. This suggests that clustering is not an appropriate approach for this dataset. Therefore, our suggestion would be to use a non-clustering approach when possible and, if not, probably the fuzzy clustering will be the recommended one.

In summary, the results show that while clustering algorithms can effectively identify clusters in numerical datasets, they struggle with mixed and categorical datasets. For all datasets, the correct number of clusters (matching the unique ground truth labels) was successfully identified. However, when dealing with categorical or binary data, the clustering process did not always produce accurate or well-formed clusters. This suggests that alternative clustering methods, such as K-Prototypes, K-Modes, or probabilistic clustering algorithms, may be better suited for these types of data.

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